

2025 Jan Oppe 1 - Set 2

Section 1 - problem 3

Find missing number in a range of numbers

PROBLEM STATEMENT

Given a list with numbers from 1 to n (inclusive) except one number in the range, write a function to find the missing number.

Assume that only one number from the range will be missing in the given list.



EXAMPLE & EXPLANATION

Given the inputs `nums=[1, 4, 2, 4, 6, 5, 6]` and `n` is 6.

Explanation:

- The number 3 is missing from the range. So, `find_missing_number(nums)` should return 3 as it is the missing number.

THE PROBLEM SPECIFICATION

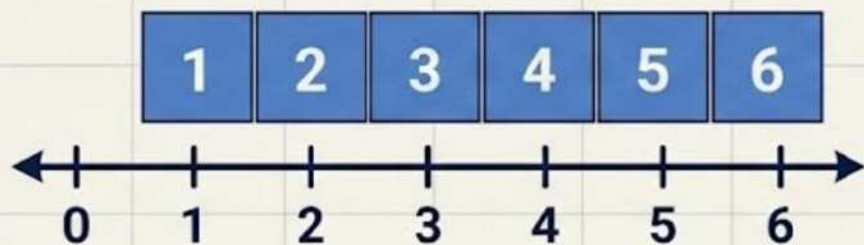


Note: This is a function type question, you don't have to take input or print the output, just have to complete the required function definition.

Visualizing the Missing Number

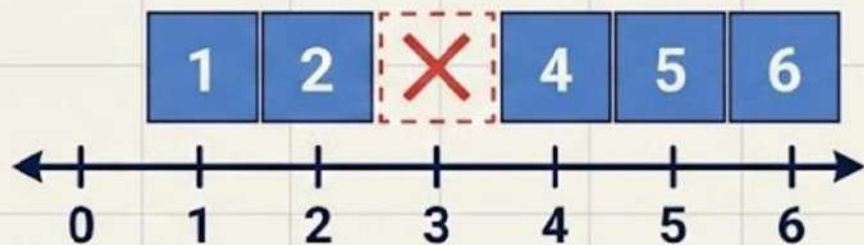
Example Data: `nums = [1, 4, 2, 4, 6, 5, 6]`, $n = 6$

Expected Range (1 to $n=6$)



All numbers from 1 to 6 should be present

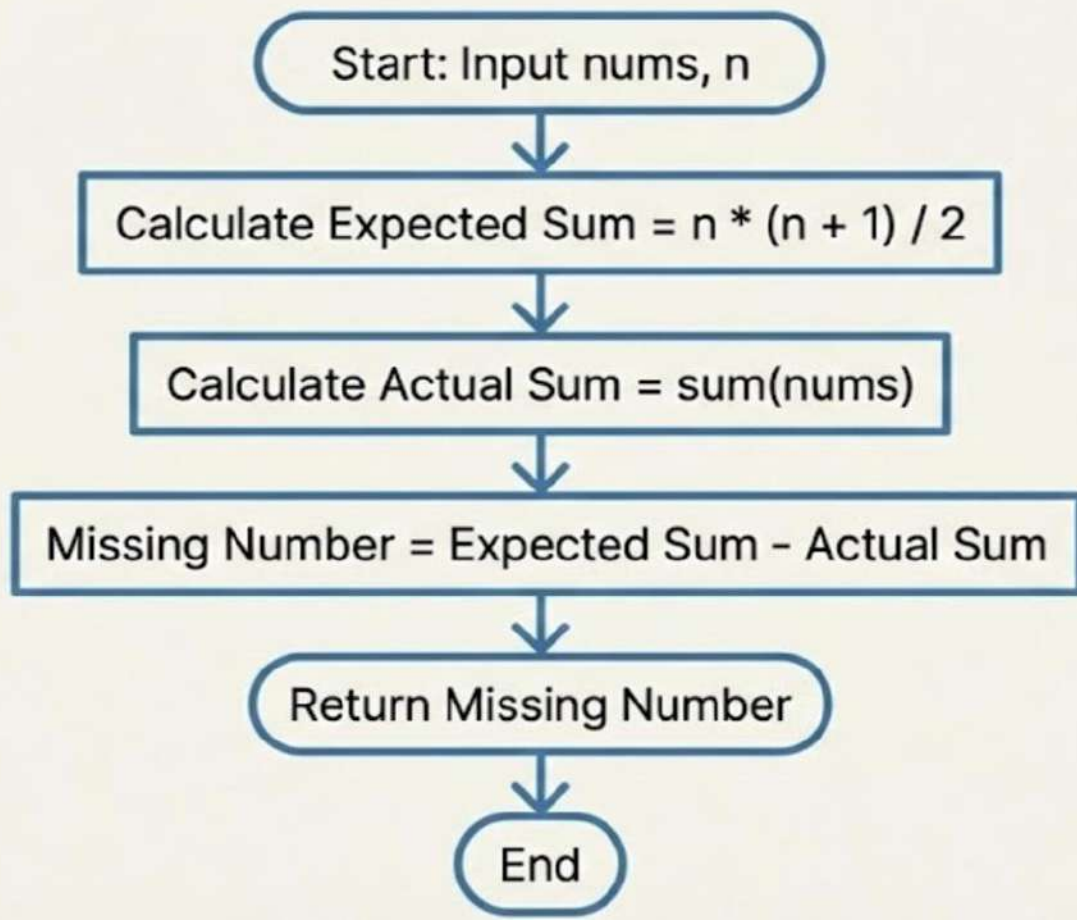
Actual List (`nums`)



Number 3 is missing from the list

Result: Missing Number is 3

The Logic Flowchart



The Explanatory Approach

Step-by-Step implementation for clarity.

```
def find_missing_number(nums, n):
```

```
# Step 1: Calculate the expected sum of  
numbers from 1 to n
```

```
expected_sum = n * (n + 1) // 2
```

Formula for the sum of
the first 'n' integers.

```
# Step 2: Calculate the sum of numbers in  
actual_sum = sum(nums)
```

Calculates the sum of
the given list.

```
# Step 3: The missing number is the differ  
missing_number = expected_sum - actual_sum
```

The difference reveals
the missing value.

```
return missing_number
```

Returns the final integer
result.

The Pythonic Optimization

Leveraging built-ins for concise logic.

Type Hinting
for clarity.

```
def find_missing_number(nums: list, n: int) -> int:  
    return (n * (n + 1) // 2) - sum(nums)
```

Arithmetic &
Integer Division
Operators.

Efficient one-liner using math
formula & built-in sum().

Result: Identical output, cleaner syntax.

Concept Spotlight: Sum of First 'n' Numbers

A clever way to count without adding everything up.

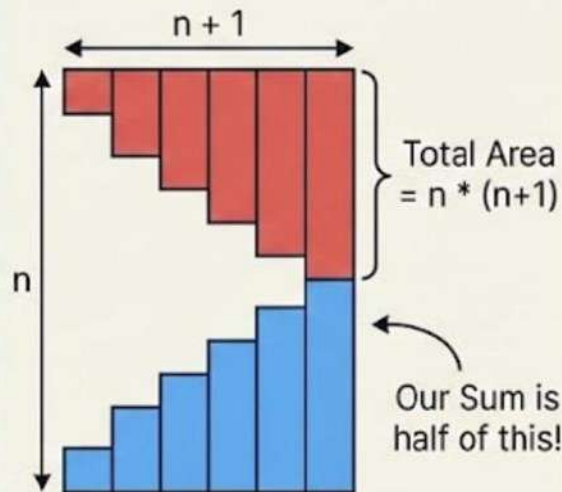
Visual Logic (The Staircase Method)

Imagine the numbers 1 to 'n' as a staircase of blocks. If you take a second, identical staircase and flip it upside down, you can fit them together to form a perfect rectangle. The area of this rectangle is easy to calculate.

The Formula

$$\text{Sum} = n * (n + 1) // 2$$

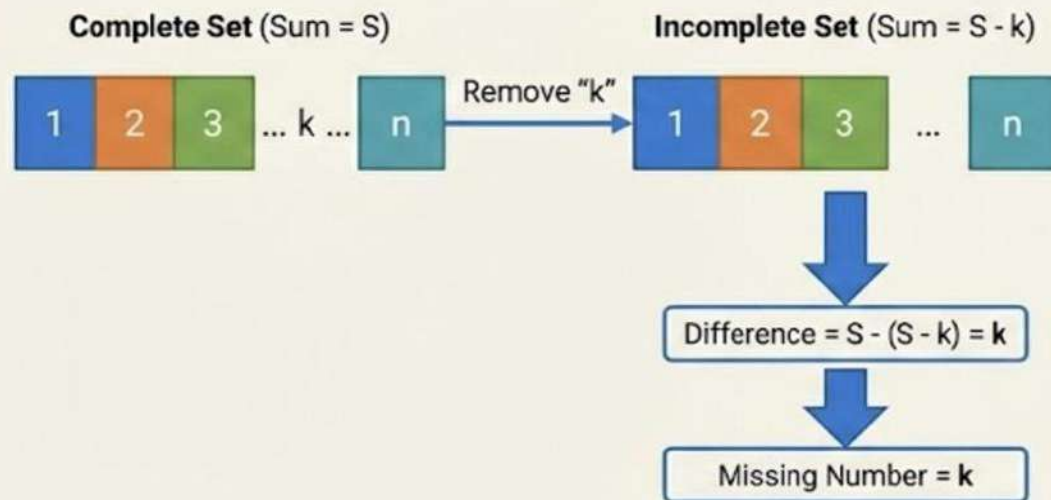
Visual Proof (n=6)



Application: For $n = 6$, the sum is $6 * (6 + 1) // 2 = 6 * 7 // 2 = 42 // 2 = 21$.

Concept Spotlight: Subtraction for Missing Value

Finding a missing number by comparing expected and actual sums.



Why Subtraction Works: If one number is missing, the total sum becomes smaller. The difference gives exactly the missing number.

Scenario A: $n=5$, list=[1, 2, 4, 5]
Expected Sum (1 to 5): $5*(6)/2 = 15$
Actual Sum: $1+2+4+5 = 12$
Difference: $15 - 12 = 3$. Missing: 3 ✓

Scenario B: $n=4$, list=[2, 3, 4]
Expected Sum (1 to 4): $4*(5)/2 = 10$
Actual Sum: $2+3+4 = 9$
Difference: $10 - 9 = 1$. Missing: 1 ✓

Scenario C: $n=6$, list=[1, 2, 3, 4, 5]
Expected Sum (1 to 6): $6*(7)/2 = 21$
Actual Sum: $1+2+3+4+5 = 15$
Difference: $21 - 15 = 6$. Missing: 6 ✓

Robustness Rule: Always validate that the input list contains unique integers within the specified 1 to n range before applying this logic.

Solution Comparison



Step-by-Step Approach

- Easy to read and understand.
- Uses intermediate variables for clarity.

```
def find_missing_number(nums, n):  
    # Calculate the expected sum of 1 to n  
    expected_sum = n * (n + 1) // 2  
    # Calculate the sum of the given list  
    actual_sum = sum(nums)  
    # The difference is the missing number  
    return expected_sum - actual_sum
```



Concise One-Liner

- More efficient and compact.
- Directly returns the calculation.

```
def find_missing_number(nums, n):  
    # Directly return the difference of  
    # expected and actual sums  
    return (n * (n + 1) // 2) - sum(nums)
```

The solution uses the arithmetic series formula to find the expected sum and subtracts the actual sum to find the missing number.

Key Takeaways

01

Problem: Find Missing Number

Write a function to find the single missing number in a list containing integers from 1 to n.

02

Core Logic: Sum Difference

Calculate the expected sum of 1 to n using a formula. Subtract the actual sum of the list to find the missing value.

03

Return Value: Integer

The function must return a single integer which is the missing number.

Final Syntax Cheat Sheet

```
return (n * (n + 1) // 2) - sum(nums)
```